Bi-Modal Transformer-Based Approach for Visual Question Answering in Remote Sensing Imagery

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***Abstract*— Recently, vision-language models based on transformers are gaining popularity for joint modeling of visual and textual modalities. In particular, they show impressive results when transferred to several downstream tasks such as zero and few-shot classification. In this article, we propose a visual question answering (VQA) approach for remote sensing images based on these models. The VQA task attempts to provide answers to image-related questions. In contrast, VQA has gained popularity in computer vision, in remote sensing, it is not widespread. First, we use the contrastive language image pretraining (CLIP) network for embedding the image patches and question words into a sequence of visual and textual representations. Then, we learn attention mechanisms to capture the intradependencies and interdependencies within and between these representations. Afterward, we generate the final answer by averaging the predictions of two classifiers mounted on the top of the resulting contextual representations. In the experiments, we study the performance of the proposed approach on two datasets acquired with Sentinel-2 and aerial sensors. In particular, we demonstrate that our approach can achieve better results with reduced training size compared with the recent state-of-the-art.**

***Index Terms*— Co-attention, remote sensing, self-attention, transformers, vision-language models, visual question answering (VQA).**

I. Introduction

REMOTE sensing technology has undergone steady progress very recently. It powers a myriad of civilian and military applications, thanks to the deployment of quantitative and qualitative sensors, on the one hand, and the evolution of powerful hardware and software platforms, on the other hand.

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Remote sensing image analysis has been acquiring a particular interest, owing to the amount of data potentially present in an image, which enables many tasks such as land use/land cover classification [1], [2] and object detection [3], [4]. In this regard, effective image analysis often remains dependent on the pinpointing of objects therein. Early systems relied on shallow representations which demonstrated rather limited performance. The appearance of deep learning strategies has brought several breakthroughs in advancing RS image understanding.

To provide a comprehensive human-like interpretation of the RS information, it is crucial to integrate natural language processing (NLP) with visual understanding to go from merely recognizing objects present in the scene to describe the attributes of the objects and the relation between them. To this end, joint vision-language modeling has been introduced recently for the tasks of describing the content of images via captioning [5]–[8] and image retrieval using textual queries [9], [10].

Beyond image captioning and text-to-image retrieval, recent literature suggests another query fashion, namely visual question answering (VQA) [11], which consists of providing feedback about an image-relevant question. Therefore, unlike prior image retrieval and captioning tasks, in VQA, both the image and a specific question are given as inputs (see Fig. 1). Yet, with respect to image captioning which normally generates a holistic natural language description of the image, VQA offers the advantage of retrieving finer details (e.g., the specifics of certain objects in the image such as the area they occupy within the image, the relative position of an object with respect to another object, and the density of certain objects across a particular area in the image, among others). Therefore, it is rational to regard VQA as a complementary component to captioning.

A typical VQA pipeline involves four modules, namely: 1) a feature extractor to draw meaningful visual cues from the query image; 2) a textual feature extractor from the query question; 3) an embedding component to combine the earlier two feature modalities; and 4) a prediction head [12]. In the context of RS imagery, the few existing VQA models use long short-term memory networks (LSTMs) for question modeling. Yet in recent years, another type of model called transformer has revolutionized the field of NLP, pushing the state-of-the-art for several NLP tasks. Furthermore, the emergence of joint

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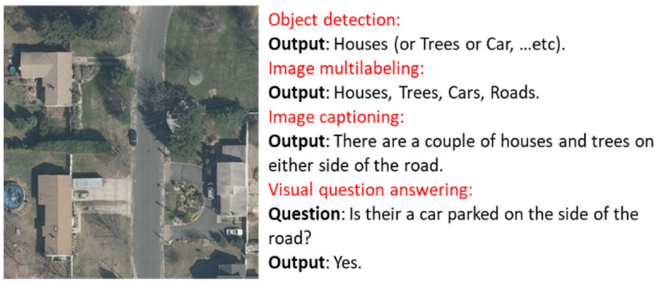


Fig. 1. Object-level remote sensing image analysis approaches.

vision-language modeling based on transformers also showed to be a promising alternative for representation learning compared with standard paradigms that learn to map images to discrete labels. Instead, they learn to align images to raw text such as contrastive language image pretraining (CLIP) [13] and object-semantics aligned pretraining (OSCAR) [14]. This type of learning allows transferring knowledge to a boarder range of downstream tasks such as classification, zero shot, and few-shot learning to name a few.

In this work, we exploit these models for transferring knowledge to our VQA task. We use the CLIP network, based on vision and language transformer encoders, as feature extractor for embedding the image patches and question words into sequence of feature representations. Then, we learn attention mechanisms through two decoders to capture the intradependencies and interdependencies between these representations. In order to generate the final answer, we take the average prediction of two classifiers mounted on the top of the resulting contextual representations. The parameters of the decoders and classifiers are optimized using the well-known cross-entropy loss, while those of CLIP are kept fixed. The main contributions of this article can be summarized as follows.

1) We propose a VQA approach based on visual-language transformers models;
2) We capture the intradependencies and interdependencies within-and-between visual and textual representations using self-attention and co-attention mechanisms;
3) We show experimentally that our approach can yield promising results on low-resolution and high-resolution datasets acquired by Sentinel-2 and Aerial sensors. In particular, we show that our approach can obtain competitive results using only 10% of the original training set.

The rest of this article is organized as follows. Section II presents the related works. Section III describes the proposed VQA approach. Section IV presents the experimental results. Finally, Section V draws the conclusions and suggestions for future developments.

II. Related Works

In the following sections, we present a review of some notable works in VQA in the context of computer vision and remote sensing followed by the utilization of transformers for solving other remote sensing tasks.

A. VQA in Computer Vision

The literature suggests several VQA works in computer vision, thanks to the existing abundant large-scale datasets and rich object annotations. Respectively, only a few contributions have been developed so far in remote sensing, owing mainly to the scarcity of diverse and rich ground truths. For instance, among the notable works proposed in the general computer vision literature, one can find the work of [11], which poses the problem of open-ended VQA by prompting selective questions about the image at hand. To this end, a dataset that collects roughly 0.25-M images and 0.76-M respective questions, and 10-M answers was built. The questions ranged from a holistic style such as "how many bikes are there?" or "what kind of store is this?" to an action-associated style such as "what is the woman reaching for?" or "can you park here?" or "why is the boy on the right freaking out?" The work in [15] attempts to associate image regions to the query text by concatenating the question and answer representations, and combining them with weighted image region features. Such an approach has proven efficient to tell the network where to look in the image given a question. The work in [16] pays attention not only to the relevant image regions but also to the words that compose the posed question. Thus, a hierarchical visual-textual pipeline is presented by applying co-attention on both the image and the question, where word level, phrase level, and question level embeddings are modeled. In [31], VQA was addressed within an adversarial learning approach in order to augment training examples without altering the visual and semantic properties of ground-truth data.

Both question and image cues are essential in VQA. The work in [17] supplicates this information pair with cues from the answer itself, where image and question features are fused and branched out into two heads which are availed to: 1) predict the answer and 2) reattend image objects under guidance of answer representations, respectively.

Recent trends consider scenarios in which a question and a set of images are presented at the input, where the query regards the relations of objects or image portions within the image set, or about a topic/scene relevant to the image set [18]. Video-based VQA is another promising trend [19].

B. VQA Remote Sensing

VQA in remote sensing images, however, is still at its debut and the amount of contributions is sharply scarce. For instance, in [20], low- and high-resolution datasets were constructed by leveraging OpenStreenMap (containing geo-localized information provided by volunteers) to release questions and answers relevant to the images. Convolutional neural networks (CNNs) and LSTM networks were adopted to model the visual and textual cues, respectively. The resulting textual and visual features are fused using point-wise multiplication. In [21], convolutional features and word vectors were used to capture the visual and natural language contents, respectively. Mutual attention is adopted to improve the alignment between the two feature modalities, topped with a fully connected (FC) layer to output the answer. In another work, a large VQA dataset was built and released [22]. In particular, question

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generation is based on image labels explicitly and proceeds by: 1) composing yes/no queries about the presence of one or more land cover classes and 2) composing which/what questions regarding land cover classes.

C. Transformers for RS Tasks

Transformers are now regarded as *de facto* standard in NLP tasks. Recently, they have been extended to vision tasks yielding the so-called ViT network [23]. In the context of remote sensing, some notable works have been developed based on these backbones. For example, Bashmal *et al.* [24] present a multilabel classification method for UAV images. Bazi *et al.* [25] proposed an extensive evaluation of ViT [25] for scene classification. In another work, He *et al.* [26] introduced a hyperspectral image classification model consisting of spectral and spatial transformer blocks. Chen *et al.* [27] introduced a transformer-based change detection method in order to model contexts within the bi-temporal image which enhances the recognition of objects of interest. In another work [28], the transformer model has been applied for crop circle detection. In the context of image captioning, Shen *et al.* [29], [30] used the transformer as a replacement to LSTM. Both works confirm its superiority compared with LSTM for sequences modeling.

It can be seen that recent literature regards transformers as an efficient tool for image modeling thanks to the attention mechanism which allows them to model inter region relations.

III. Method

Let us consider a training set $D_{\text{tr}} = \{X_i, q_i, y_i\}_{i=1}^{N_{\text{tr}}}$ composed of N_{tr} input image and question pairs (X_i, q_i) associated with the desired categorical answer $y_i \in \mathcal{R}^C$ where C represents the number of possible answers. In this work, we consider closed-ended VQA, where the answers are given in the form of "yes," "no," object counting, area covered by particular objects, urban, and rural. Our aim is to learn a model that can predict answers for a set of unseen N_{ts} test image and question pairs $D_{\text{ts}} = \{X_j, q_j\}_{j=1}^{N_{\text{ts}}}$.

Thus, this VQA setting can be regarded as a multiclass classification problem with bimodal heterogonous inputs coming from vision and NLP worlds. In the following sections, we provide detailed descriptions of our VQA model shown in Fig. 2.

A. Feature Extraction Using CLIP

A common practice for learning visual representations is to use vision models pretrained on large scale datasets such as ImageNet dataset. Yet, our task involves learning from both images and texts modalities. For such purpose, we propose to explore vision-language pretrained models as an alternative solution. In particular, we adopt the CLIP model that is based on dual transformers pretrained on matching 400-M image-text pairs through contrastive loss optimization.

The text encoder is a Bert-like transformer architecture [31]. The process starts by tokenizing the words of the question q_i appended with two special tokens [SOS] and [EOS] representing the start and end of the sequence into unique numeric numbers. The vocabulary size is equal to 49 408. To facilitate batch processing, the text encoder uses a sequence with a fixed length equal to 77 in addition to a word embedding layer to embed the sequence into features of dimension 512. To supply the sequence with information about the order of each word, a learnable positional embedding is added. The resulting initial representation is then fed as input through multiple identical layers (i.e., 12 layers) to generate the final representation.

Each layer of this encoder, as shown in Fig. 3, uses a multihead self-attention (MSA) block followed by a small multilayer perceptron (MLP) block. Both blocks are connected with skip connections and use LayerNorm (LN) [32]. Compared with batch normalization (BN), LN aims at estimating the normalization statistics from the summed inputs to the neurons within a hidden layer. It has been applied in recurrent networks as a solution for improving the generalization ability. Lately, it has been adopted in transformers. The MLP is a simple feed-forward network consisting of two FC layers with Gaussian error linear unit (GELU) activation function applied in between [33]. The GELU activation function is expressed as follows:

$$\text{GELU}(x) = x\Phi(x) \tag{1}$$

where $\Phi(x)$ is the standard Gaussian cumulative distribution function. The choice of this function was made on the basis of prior NLP literature where it has been found that it handles better the problem of vanishing gradient compared with other activations. The main goal of the MSA block is to model the long-range dependencies between a specific token and all other tokens in the sequence. Each MSA block is composed of several scaled dot-product attention mechanisms. The scaled dot-product attention function is performed in parallel by multiple heads to enable the model to pay attention to more than one feature at a time. The results of all heads are concatenated and linearly projected into the desired dimension using a weighted matrix.

Finally, a linear layer is used to map the features to the dimension 512. To meet the requirements of our VQA task, we remove the pooling operation to keep the representation of each token in the sentence, thus resulting in a sequence of dimension 77×512.

On the other side, the image encoder is based on the vision transformer proposed by Dosovitskiy *et al.* [23] termed as ViT16. First, the image X_i is resized to the dimension $224 \times 224 \times 3$ pixels. Then, it is divided into 196 nonoverlapping patches of spatial dimension 16×16 pixels. These patches are flattened and mapped using a linear embedding layer. Positional encoding is appended to the patch representations. The resulting embedded sequence of patches is then fed to an encoder similar to the one used previously for NLP. As done for the NLP transformer, we discard the pooling operation to keep the sequence of visual representations of dimension 196×512 where each token of dimension 512 represents a particular region of the image of dimension 16×16 pixels.

B. Learning Contextual Representations

Let us consider $\{F_{X_i}, F_{q_i}\}_{i=1}^{N_{\text{tr}}}$ as the visual and textual representations obtained from CLIP, with $F_{X_i} \in \mathcal{R}^{1\times 196\times 512}$ and

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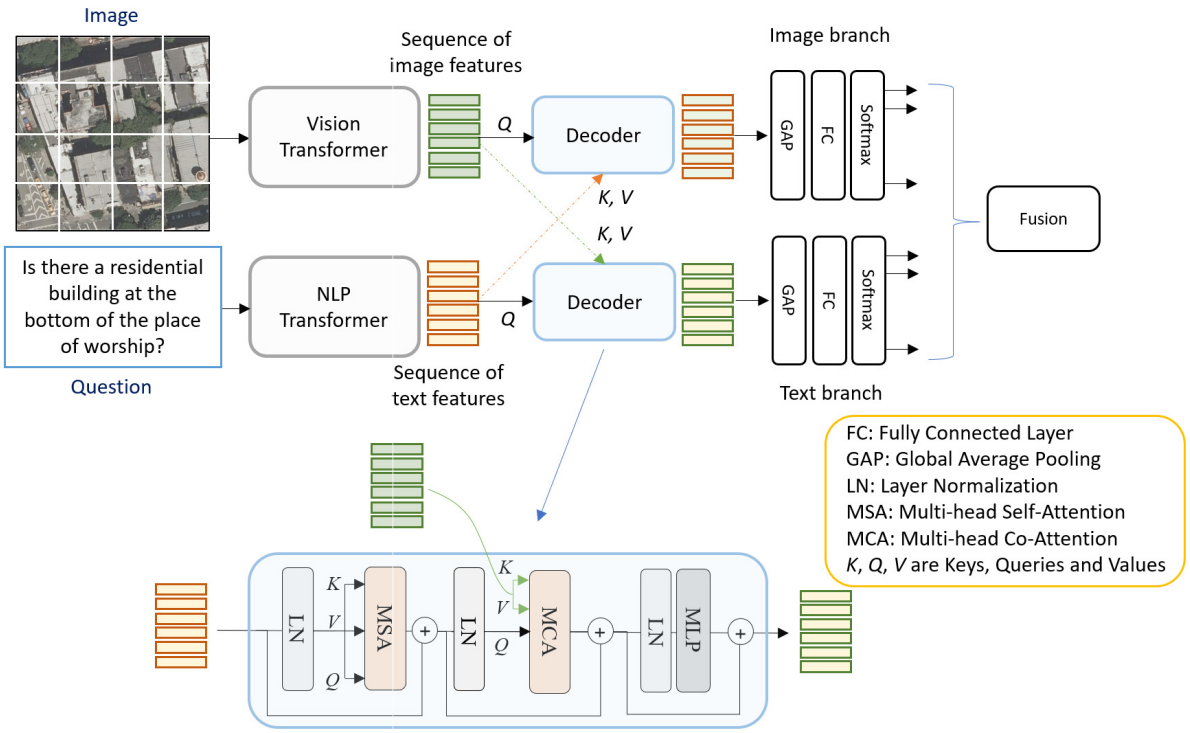


Fig. 2. Proposed VQA approach: we use a vision-language transformer to encode image and question pairs into textual and visual feature representations. Then, we use two transformer decoders to discover the contextual relationship between these representations. For classification, we average the outputs of two classifiers mounted on the top of the attended textual and visual representations.

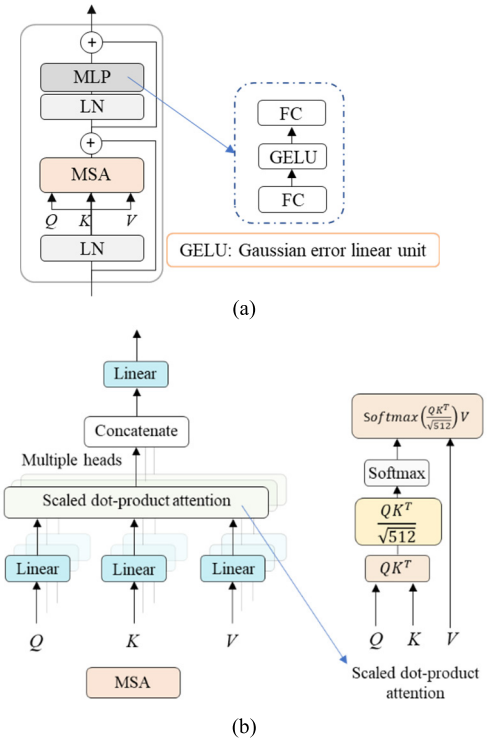


Fig. 3. (a) Self-attention layer used in the transformer with its (b) MSA block. The MSA block is composed of multiple scaled dot-product attention.

$F_{q_i} \mathcal{R}^{1\times 77\times 512}$. As mentioned previously, 196 and 77 represent the length of the visual and textual sequences, while 512 represent the feature dimension. We propose to capture from these visual and textual representations the internal dependencies among each modality as well as the cross-modality correlation. For such purpose, we use two decoders with similar architecture for generating these contextual representations. For consistency, we use two decoders with the same architecture as the blocks adopted in CLIP yet our decoders contain additional layers for co-attention. For example, we show in Fig. 2 the decoder part related to the text branch.

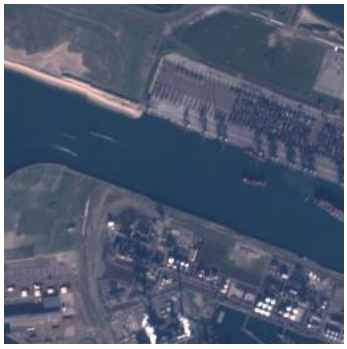
It is also composed of MSA for capturing the intradependencies among the textual representation followed by multihead co-attention (MCA) for capturing the correlation with the visual representations. Here also, both layers are connected with skip connections and use LN. The MSA layer receives F_{q_i} and projects it using the weight matrices W^Q, W^K, and $W^V \in \mathbb{R}^{512\times 512}$, such that $Q = W^Q F_{q_i}$, $K = W^K F_{q_i}$, and $V = W^V F_{q_i}$.

The MSA comprises eight independent self-attention heads operating in parallel, and each head computes a different attention score using the scaled dot-product similarity between Q, K, and V

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\text{T}}}{\sqrt{512}}\right)V. \tag{2}$$

Then, the outputs of all heads are concatenated and projected with another learnable weights matrix. The MCA block works in the same way as MSA except the key and values are now coming from the visual features: $Q = W^Q F_{q_i}$, $K = W^K F_{X_i}$, and $V = W^V F_{X_i}$. At the output of the text decoder, we obtain the conditional textual representations referred as F'_{q_i}. On the other side, the image decoder yields the conditional visual representations F'_{X_i} by considering the visual representation as queries and the textual representations as keys and values.

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Question: Is there a water area?
Output: yes: presence
Question : What is the number of parkings?
Output : 7:count
Question : Is it a rural or an urban area?
Output : urban: rural urban
Question : Are there more water areas than roads?
Output : no:comp

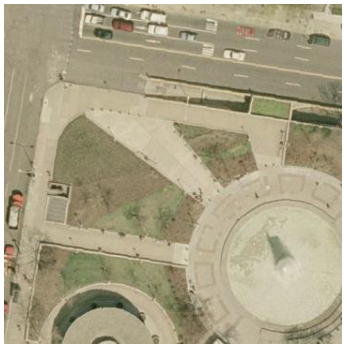


Question: Is there a building?
Output: yes: presence
Question: What is the amount of farmlands?
Output: 46:count
Question: Is it a rural or an urban area?
Output: urban: rural urban
Question: Are there less roads than forests?
Output: no:comp



Question: Is a square road present?
Output: no: presence
Question: How many piers are there?
Output: 5:count
Question: Is it a rural or an urban area?
Output: urban: rural urban
Question: Are there less forests than water areas?
Output: no:comp

(a)



Question: Is there a park?
Output: yes: presence
Question: How many roads are there?
Output : 13:count
Question : Are there more roads than buildings?
Output : yes:comp



Question: Is a commercial building present?
Output: yes: presence
Question: How many buildings are there?
Output: 3:count
Question: What is the area covered by buildings?:
Output: 5063m2 :area



Question: Is there a residential area?:
Output: yes: presence
Question: What is the amount of residential areas?
Output: 1:count
Question: What is the area covered by residential areas?
Output: 9196m2:area

(b)

Fig. 4. Example of images with different type of question answers. (a) LR and (b) HR datasets.

C. Classification

For obtaining the answer, we use one classifier per modality. For each modality, we use global average pooling (GAP) to convert the sequence into a single feature of dimension 512. Then, we feed this feature to an FC layer with Softmax activation function. Thus, the outputs of the text and image classifiers are given as follows:

$$\begin{cases} \hat{y}_i^q = \text{Softmax}\big(\text{FC}\big(\text{GAP}\big(F'_{q_i}\big)\big)\big) \\ \hat{y}_i^X = \text{Softmax}\big(\text{FC}\big(\text{GAP}\big(F'_{X_i}\big)\big)\big). \end{cases} \tag{3}$$

Then, we train the network by optimizing the standard cross-entropy loss in a batch-wise manner over the training set. The total loss over the training set is given as follows:

$$\mathcal{L} = \frac{\lambda_1}{N_{tr}} \sum\nolimits_{i=1}^{N_{\text{tr}}} \text{CrossEntropy}\big(y_i, \hat{y}_i^q\big) + \frac{\lambda_2}{N_{tr}} \sum\nolimits_{i=1}^{N_{\text{tr}}} \text{CrossEntropy}\big(y_i, \hat{y}_i^X\big) \tag{4}$$

where λ_1 and λ_2 are the two regularization parameters controlling the contributions of both losses. In our experiments, we set them to 0.5. In the prediction phase, we take the average prediction of both classifiers

$$\hat{y}_i = 0.5\big(\hat{y}_i^q + \hat{y}_i^X\big). \tag{5}$$

IV. Experimental Results

A. Dataset Description

1) Low Resolution: This dataset consists of nine different tiles of 10 m of spatial resolution, covering 6.55 km^2, which are based on Sentinel-2 images acquired over The Netherlands (RGB bands). These tiles were divided into 772 images of size 256 × 256 pixels. Then, 77 232 question–answer pairs were generated. For each image, many questions–answers were given (around 100 question–answer per image). The questions are mainly related to object presence (answer: yes/no), comparisons between objects (answer: yes/no), rural or urban areas, and object counting. Regarding object counting, it was proposed in [20] to assign object counts in specific ranges (answer: equal to 0; greater than 0 and smaller or equal to 10; greater than 10 and smaller or equal to 100; greater

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than 100 and smaller or equal to 1000; and greater than 1000). Thus, the maximum number of possible answers for this dataset is $C = 11$. We follow the split proposed in [20] by taking 77.8% of the original tiles, 11.1% and 11.1%, for training, validation, and test, respectively. In Fig. 4(a), we depict examples of images, questions, and answers. In contrast, in Fig. 5(a) and (b), we show the distribution of the answers for this dataset for both train and test sets.

2) High Resolution: This dataset contains 161 different tiles of aerial images with 15 cm of spatial resolution. These tiles were divided into 10 659 images of size 512 × 512 pixels, where each image covers covering 5898 m^2. Similarly, 1 066 316 question–answer pairs were generated (around 100 question–answer per image). The questions are related to object counting (answer: from 0 to 89 object counts), presence (answer: yes/no), comparison (answer: yes/no), and area covered by particular objects in specify ranges similar to object counting for the LR dataset (answer: equal to 0; greater than 0 and smaller or equal to 10; greater than 10 and smaller or equal to 100; greater than 100 and smaller or equal to 1000; and greater than 1000). Based on this division, the number for classes in this dataset is $C = 99$. The training set holds 61.5% of the tiles, 11.2% are allocated to the validation set, and the rest as test sets (20.5% for test set 1 and 6.8% for test set 2). Here, we notice that test set 2 is used for assessing the robustness of the model on image acquired at different locations. In Fig. 4(b), we show the examples of images, questions, and answers for this HR dataset. As done for the LR dataset, we show in Fig. 5(c) and (d), the distribution of the answers and examples of image–question pairs. For ease of interpretation, we present the number of counting answers as a single class yet in the experiments, we consider 89 categorical counting classes as mentioned previously. It is worth recalling that the images of Fig. 4 were selected randomly so as to cover a variety of question types (count, presence, comparison, and urban/rural) for the LR dataset and (count, presence, comparison, and area) for the HR dataset.

B. Experimental Setup and Evaluation Metrics

We implement our method using PyTorch library [34] on a PC with Core(TM) i9-7920X CPU, GeForce RTX 1080 Ti 11 GB, and 64-GB RAM.

We optimize our model using Adam optimizer [35] with a learning rate of 0.0001 and for 30 iterations. We set the minibatch size to 120. For data augmentation, we resize the images to 288 × 288 pixels and then we randomly crop 224 × 224 pixels. Then, we apply also random rotation by 20°.

For the LR dataset, we present the accuracies per question type: count, presence, comparison, and urban/rural. In contrast, for the HR dataset, we group them into count, presence, comparison, and area.

C. Results on LR Dataset

We train the model on the training set for three different trials with different initializations. Then, we present the averaged classification accuracies for these three trials along with the standard deviation. From Table I, we see that the text classifier (with the output $\hat{y}_i^q$) gives an average accuracy (AA) and overall accuracy (OA) of (85.96% and 84.76%). While the image classifier (with the output $\hat{y}_i^X$) performs slightly better, the respective scores are (86.14% and 84.73%). In contrast, fusing both classifiers (the output $\hat{y}_i$) leads to (86.78% and 85.56%). We can observe that when the question type is relevant to presence, comparison, and urban/rural categories, the scores exceed 91%. Yet, for the counting class, it is only 72.22%, which indicates that this class is more challenging as we have to assign objects in specific ranges. Comparing our results to the work presented in [20], we observe significant improvements for all classes with 2.66% up to 9.66% for urban/rural and count classes, respectively. The increase in (OA and AA) accuracies is (5.29% and 6.48%).

TABLE I

CLASSIFICATION RESULTS OBTAINED FOR THE LR TEST SET. THE NUMBERS BETWEEN BRACKETS INDICATE THE STANDARD DEVIATION

Type	Ours			Baseline
	Text	Image	Fusion	[20]
Count	70.47% (0.35%)	71.13%(0.85%)	**72.22%(0.57%)**	67.01% (0.59%)
Presence	90.64% (0.52%)	90.58%(0.53%)	**91.06%(0.17%)**	87.46% (0.06%)
Comparison	90.74%(0.17%)	90.53%(0.49%)	**91.16%(0.09%)**	81.50% (0.03%)
Urban/Rural	92.00%(1.73%)	92.33%(1.52%)	**92.66%(1.52%)**	90.00% (1.41%)
AA	85.96%(0.62%)	86.14%(0.43%)	**86.78%(0.28%)**	81.49% (0.49%)
OA	84.76%(0.33%)	84.73%(0.37%)	**85.56%(0.16%)**	79.08% (0.20%)

TABLE II

CLASSIFICATION RESULTS OBTAINED FOR THE FIRST HR TEST SET

Type	Ours			Baseline
	Text	Image	Fusion	[20]
Count	69.30%(0.12%)	69.26%(0.12%)	**69.80%(0.09%)**	68.63%(0.11%)
Presence	91.66%(0.03%)	91.65%(0.09%)	**92.03%(0.08%)**	90.43%(0.04%)
Comparison	91.01%(0.05%)	91.29%(0.08%)	**91.83%(0.00%)**	88.19%(0.08%)
Area	85.78%(0.16%)	85.72%(0.18%)	**86.27%(0.05%)**	85.24%(0.05%)
AA	84.44%(0.04%)	84.48%(0.12%)	**84.98%(0.05%)**	83.12% (0.03%)
OA	84.74%(0.04%)	84.81%(0.11%)	**85.30%(0.05%)**	83.23%(0.02%)

D. Results on HR Datasets

Table II reports the scores obtained for the first HR dataset. Here, both text and image classifiers yield close results and their fusion yields slight improvement scoring an (AA and OA) equal to (84.98% and 85.30%). The counting class is the worst one as we obtain 69.80%. Regarding the second HR test dataset, as shown in Table III, the accuracies are less compared with the first HR dataset due to the data-shift problem as the images are acquired at different locations. Here again, the text classifier performs slightly better compared with the image classifier and their fusion yields an (AA and OA) of (80.54% and 81.23%). Comparing these results to the method proposed in [20], we observe that our method performs better on both datasets.

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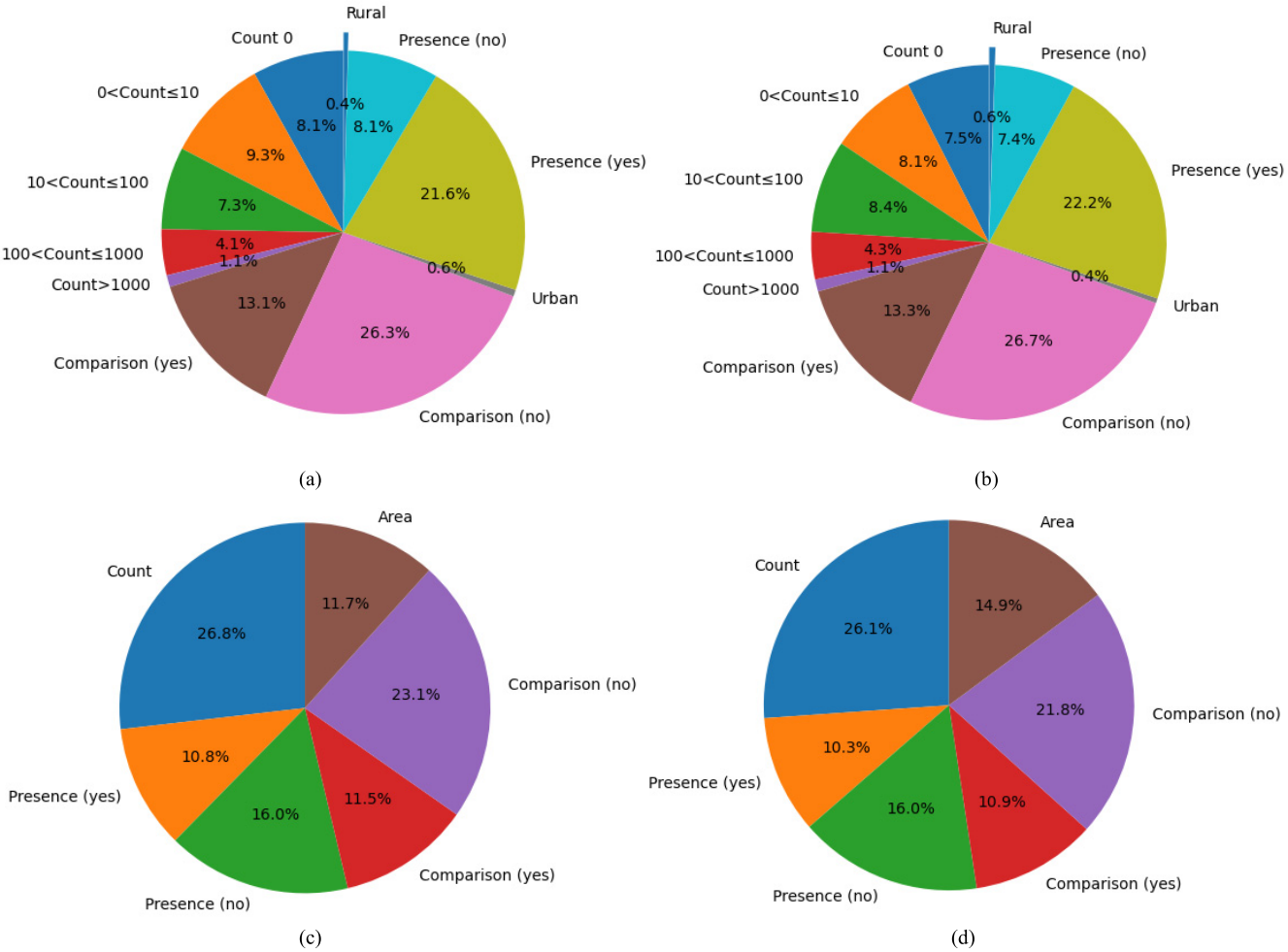


Fig. 5. Distribution of classes per question type. (a) Train and (b) test of LR dataset. (c) Train and (d) first test set of the HR dataset.

TABLE III
CLASSIFICATION RESULTS OBTAINED FOR THE SECOND HR TEST SET *t*

Type	Ours			Baseline [20]
	Text	Image	Fusion	
Count	62.97% (0.28%)	62.44%(0.11%)	**63.06%(0.11%)**	61.47%(0.08%)
Presence	88.74% (0.28%)	88.86%(0.21%)	**89.37%(0.21%)**	86.26% (0.47%)
Comparison	88.82%(0.55%)	88.34%(0.22%)	**89.62%(0.29%)**	85.94% (0.12%)
Area	79.55%(0.70%)	79.29%(0.63%)	**80.12%(0.30%)**	76.33% (0.50%)
AA	80.02%(0.44%)	79.73%(0.18%)	**80.54%(0.16%)**	77.50% (0.29%)
OA	80.69%(0.42%)	80.39%(0.11%)	**81.23%(0.15%)**	78.23% (0.25%)

E. Sensitivity With Respect to the Training Set Size

As mentioned previously, around 100 questions were generated automatically for each image for both LR and HR datasets. In this experiment, we analyze the sensitivity of the model with respect to low regime training size scenarios. We repeat the above experiments using 20% and 10% of the training sets by sampling over questions. Table IV shows the classification results expressed in terms of OA and standard deviation for the LR dataset for three trials with different training subsets. Our model yields an (AA and OA) of (85.72% and 84.30%) and (83.83% and 82.85%) for the scenarios 20% and 10%, respectively, resulting a decrease of (1.06% and 1.26%) and (2.95% and 2.71%) compared with the original training set. Yet, it still provides better scores compared with [20] by using only 10% of the training set. Similar observations can be made for the HR datasets (see Tables V and VI), where the model reaches almost the same accuracies for the first HR dataset with only 10% of the training set. Furthermore, it performs better for the second HR dataset with 10% of training set. This confirms clearly the ability of the network in learning suitable visual and textual representations for the VQA task.

F. Impact of Decoders

As mentioned previously, we append two transformer decoders to model the contextual relationship between the visual and textual image representations. To this end, we carry out experiments by removing this part and assess the performance of the model. The results are reported in Tables VII and VIII for LR and HR datasets, respectively. Evidently, discarding the decoder part incurs a significant drop in the performance on both datasets. For instance, considering the question of type "comparison" in the HR dataset, roughly 25% drop is observed when the decoder is discarded, and an

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TABLE IV

CLASSIFICATION RESULTS OBTAINED FOR THE LR TEST SET USING 20% AND 10% OF THE TRAINING SET. THE METHOD PROPOSED IN [35] USES THE ENTIRE TRAINING SET

	Train (20%)			Train (10%)			Baseline[20]
Type	Text	Image	Fusion	Text	Image	Fusion	
Count	68.24%(0.25%)	68.63%(1.17%)	**69.98%(0.60%)**	65.58%(0.62%)	67.13%(1.07%)	67.67%(0.67%)	68.63%(0.11%)
Presence	89.17%(0.66%)	89.18%(0.51%)	89.99%(0.20%)	87.57%(0.17%)	87.97%(0.15%)	88.38%(0.25%)	**90.43%(0.04%)**
Comparison	90.11%(0.43%)	89.73%(0.73%)	**90.60%(0.54%)**	89.32%(0.44%)	89.91%(0.83%)	90.27%(0.47%)	88.19%(0.08%)
Urban/Rural	90.33%(2.51%)	90.33%(2.51%)	**92.33%(0.57%)**	87.26%(2.64%)	88.66%(0.57%)	89.00%(1.00%)	85.24%(0.05%)
AA	84.46%(0.76%)	84.55%(0.65%)	**85.72%(0.39%)**	82.37%(0.74%)	83.42%(0.43%)	83.83%(0.13%)	83.12% (0.03%)
OA	83.29%(0.33%)	83.30%(0.63%)	**84.30%(0.34%)**	81.60%(0.19%)	82.41%(0.35%)	82.85%(0.14%)	83.23%(0.02%)

TABLE V

CLASSIFICATION RESULTS OBTAINED FOR THE FIRST HR TEST SET USING 20% AND 10% OF THE TRAINING SET. THE METHOD PROPOSED IN [35] USES THE ENTIRE TRAINING SET

	Train (20%)			Train (10%)			Baseline [20]
Type	Text	Image	Fusion	Text	Image	Fusion	
Count	66.85%(0.10%)	67.29%(0.11%)	68.22%(0.13%)	66.84%(0.30%)	66.98%(0.08%)	68.04%(0.08%)	**68.63%(0.11%)**
Presence	90.45%(0.21%)	90.56%(0.08%)	**91.40%(0.05%)**	90.32%(0.10%	90.26%(0.20%)	91.21%(0.12%)	90.43%(0.04%)
Comparison	89.74%(0.20%)	89.68%(0.24%)	**90.89%(0.08%)**	89.01%(0.06%)	88.92%(0.09%)	90.31%(0.02%)	88.19%(0.08%)
Urban/Rural	83.38%(0.12%)	83.51%(0.17%)	**84.81%(0.13%)**	83.09%(0.50%)	83.45%(0.28%)	84.73%(0.39%)	85.24%(0.05%)
AA	82.61%(0.11%)	82.76%(0.06%)	**83.83%(0.03%)**	82.32%(0.10%)	82.40%(0.05%)	83.57%(0.06%)	83.12% (0.03%)
OA	83.01%(0.11%)	83.15%(0.10%)	**84.20%(0.02%)**	82.69%(0.09%)	82.73%(0.04%)	83.90%(0.02%)	83.23%(0.02%)

TABLE VI

CLASSIFICATION RESULTS OBTAINED FOR THE SECOND HR TEST SET USING 20% AND 10% OF THE TRAINING SET. THE METHOD PROPOSED IN [35] USES THE ENTIRE TRAINING SET

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	Train (20%)			Train (10%)			Baseline [20]
Type	Text	Image	Fusion	Text	Image	Fusion	
Count	61.63%(0.03%)	61.47%(0.88%)	**62.36%(0.54%)**	61.15%(0.19%)	60.71%(0.29%)	61.87%(0.29%)	61.47%(0.08%)
Presence	86.37%(2.31%)	87.49%(0.34%)	**88.22%(0.79%)**	87.58%(0.13%)	86.95%(0.54%)	88.15%(0.40%)	86.26% (0.47%)
Comparison	87.32%(0.37%)	87.12%(0.21%)	**88.81%(0.34%)**	86.39%(0.81%)	86.46%(0.43%)	88.02%(0.20%)	85.94% (0.12%)
Urban/Rural	77.28%(1.61%)	77.94%(0.30%)	78.90%(0.42%)	78.52%(0.41%)	78.13%(1.10%)	**79.50%(0.61%)**	76.33% (0.50%)
AA	78.15%(0.88%)	78.50%(0.26%)	**79.57%(0.08%)**	78.41%(0.27%)	78.06%(0.48%)	79.39%(0.27%)	77.50% (0.29%)
OA	78.89%(0.71%)	79.17%(0.26%)	**80.30%(0.01%)**	78.96%(0.29%)	78.64%(0.39%)	79.98%(0.20%)	78.23% (0.25%)

TABLE VII

CLASSIFICATION RESULTS OBTAINED FOR THE LR TEST SET WITH AND WITHOUT DECODERS

Type	Without Decoders	With Decoders
Count	53.18%	**72.22%**
Presence	69.77%	**91.06%**
Comparison	77.88%	**91.16%**
Urban/Rural	84.26%	**92.66%**
AA	71.27%	**86.78%**
OA	67.37%	**85.56%**

TABLE VIII

CLASSIFICATION RESULTS OBTAINED FOR THE HR1 AND HR2 TEST SETS WITH AND WITHOUT DECODERS

	HR1		HR2	
Type	Without Decoders	With Decoders	Without Decoders	With Decoders
Count	61.51%	**69.80%**	57.38%	**63.06%**
Presence	82.16%	**92.03%**	77.94%	**89.37%**
Comparison	66.54%	**91.83%**	66.80%	**89.62%**
Area	72.89%	**86.27%**	67.85%	**80.12%**
AA	70.77%	**84.98%**	67.50%	**80.54%**
OA	70.22%	**85.30%**	67.40%	**81.23%**

OA accuracy drop of about 18% is noticed for the LR dataset, which underlines the importance of taking intermodality feature relationship into account.

G. Attention Maps

In the following, we show the attention maps provided by the proposed bi-modal transformer model over questions and

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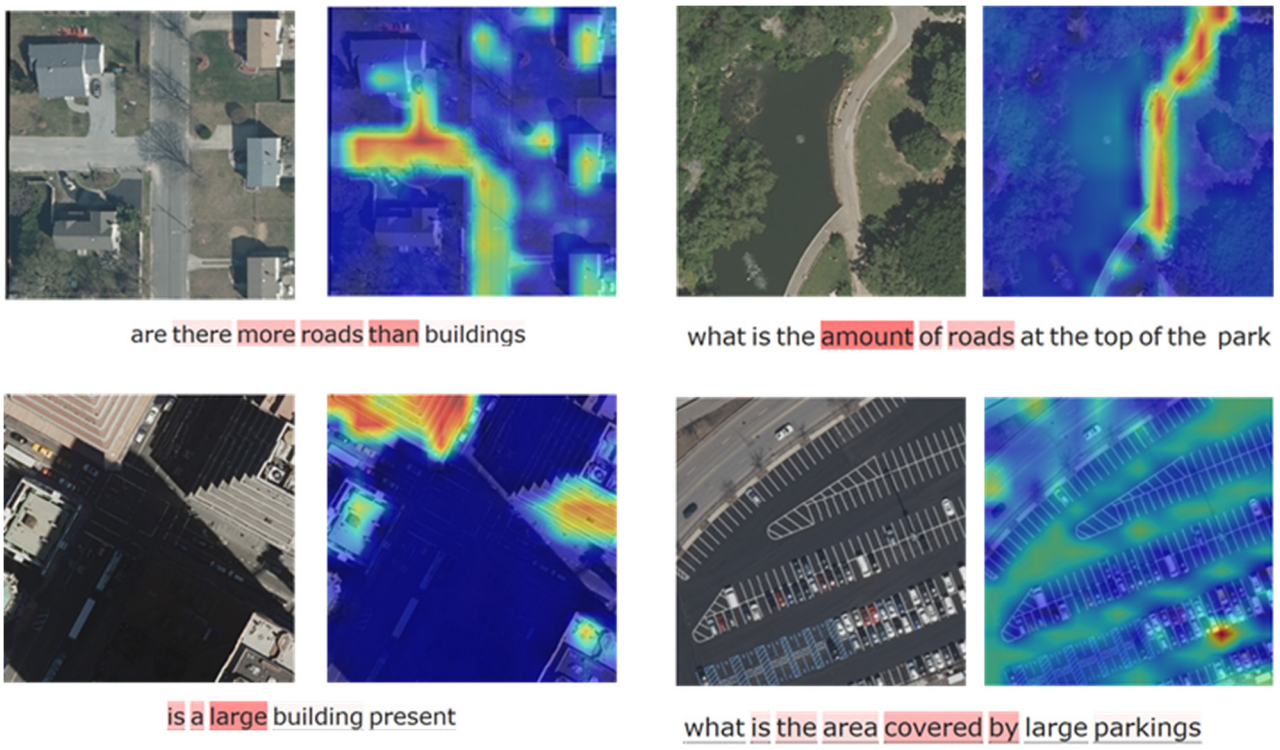


Fig. 6. Attentions over text and images for the HR dataset.

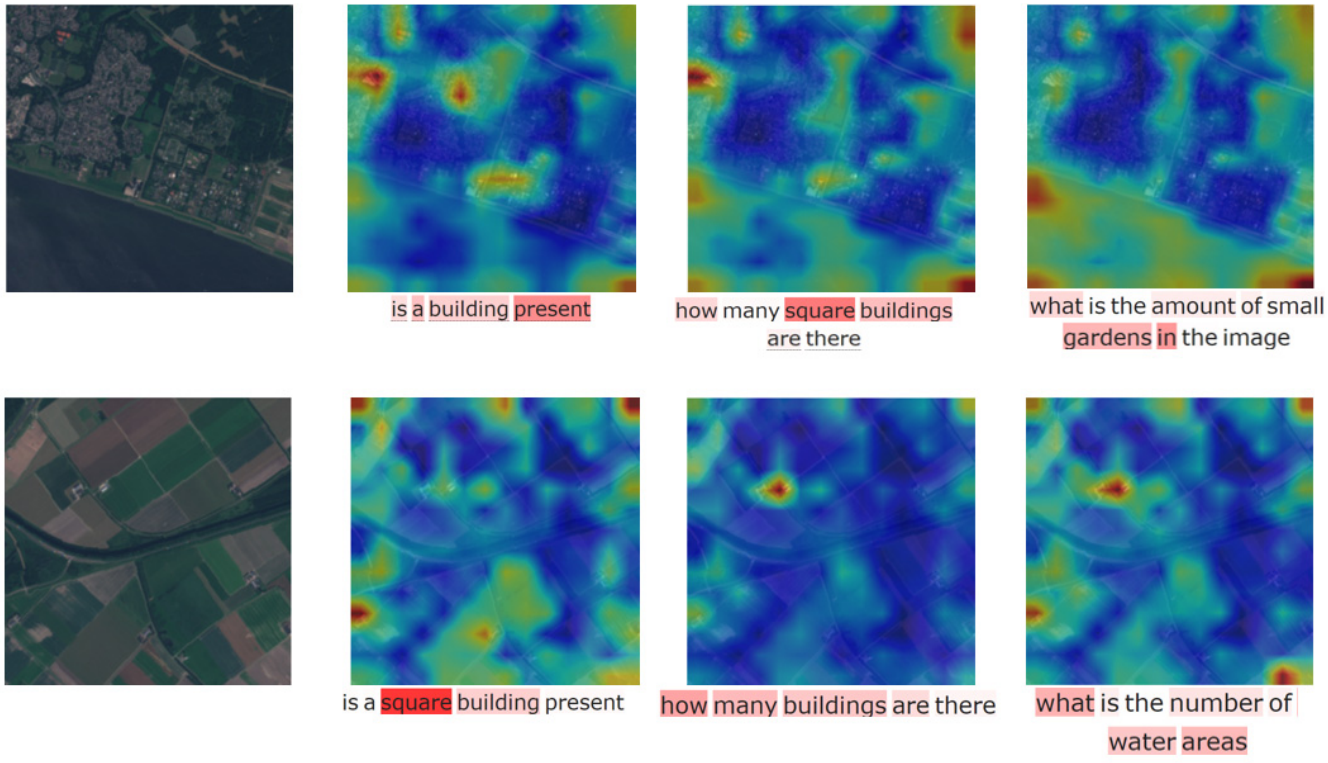


Fig. 7. Attentions over text and images for the LR dataset.

images to understand relevance of text and image regions for generating the answer. To capture the attention maps from the output token to the input space of the network, we adopt the attention rollout strategy proposed in [36]. Basically, at every transformer block, an attention matrix that quantifies attention flow from a certain token in the previous layer to another token in the next layer is envisioned. Therefore, to track down attention flow between two matrices, their respective weight matrices are recursively multiplied. As proposed in the original paper [36], we consider the average of the attention heads.

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Fig. 6 shows the attention for text and question pairs generated from HR dataset. It can be observed that when the question queries about more than one object, such as the top left example which compares the amount of "roads" with respect to "buildings," the network seems to emphasize on both of the objects as shown in the attention map. It can also be noted that the object "roads" displays higher peaks in the attention map, owing to the fact that it is the object that is stressed in the query question ("more" roads than buildings).

Interestingly, querying the amount of roads (top right example) incites the network to focus on the object "roads" in the image. The same holds for the two examples at the bottom where the network looks at the whereabouts of the objects that are present in the query questions (i.e., buildings and parkings, respectively). This suggests an evident consistency between the input question and the attention of the network.

In Fig. 7, we depict the attention maps produced from the LR dataset. For instance, in the top example, when the question queries about the object "building," the attention maps reflect relevant areas in the image, paying less attention to the bottom of the image where no buildings are observed. When the query regards the object "gardens," however, more attention is given to the bottom of the image (rightmost attention map of the top example). As per the second example, querying the presence of the "square building" object stresses diverse areas across the image although the image consists mostly of agricultural fields, which may be due to the constraint "square" that characterizes both the buildings and the fields. When querying about the number of buildings (second map), the attention drops significantly with respect to the previous case, while querying about the presence of water (last map), sparse attention can be observed. On this point, it is to note that the attention maps in the LR case suggest less relevance with the image compared with the HR case that offers major spatial details.

Although the illustrated examples show a promising coherence between the questions and the network's focus areas in particular for the HR dataset, we believe that even finer results can be obtained. In particular, it is to point out that the network does not receive the bounding box annotations of the objects included in the questions, which is an important information that can support the network in the training. In contrast, we deal with images that were acquired from a top-view, which is known to disregard the facade of the objects of interest. Another contributing factor is the construction of the dataset itself, where most of the questions were produced automatically, yet many of them can be rather noisy and could be misleading to the network.

V. Conclusion

In this article, we have introduced a VQA approach based on vision and language transformers. We have proposed to capture the dependencies between visual and textual representations using co-attention mechanisms. In the experimental results, we have shown that our approach can yield promising results in terms of classification accuracy compared with the existing solutions with reduced training set size. For future developments, we propose to enhance the method by increasing the diversity among the two classifiers. In terms of dataset, we suggest adding object-based annotations to better capture the correlation between the textual and visual cues. Another possible interesting development is to extend the dataset to open-ended VQA scenarios where the answer can be given as a natural language.

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